# EX1 – Introduction to System Programming

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## Unix ex 1 - Unix Basics

Log on a Linux machine. Enter your login (user name) and password at relevant prompts.

open terminal.

Enter these commands at the UNIX prompt. For each command copy paste the output and note its role.

- `echo hello world` ↵
- `date` ↵
- `hostname` ↵
- `arch` ↵
- `uname -a` ↵
- `uptime` ↵
- `who am i` ↵
- `who` ↵
- `id` ↵
- `last` ↵
- `w` ↵
- `top` ↵ (you may need to press `q` to quit)
- `echo {con,pre}{sent,fer}{s,ed}` ↵
- `man ls` ↵ (you may need to press `q` to quit)
- `man who` ↵ (you may need to press `q` to quit)
- `clear` ↵
- `bc -l` ↵(type `quit` ↵ or press Ctrl-d to quit)
- `echo 5+4 | bc -l` ↵
- `yes please` ↵(you may need to press Ctrl-c to quit)
- `time sleep 5` ↵
- `history` ↵
- `nano` ↵

1. **echo** – Prints the words following the "echo" command to the terminal.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ echo hello world
hello world
```

2. **date** – Print to the terminal the current date and time.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ date
Sat Mar 28 19:30:38 IDT 2026
```

3. **hostname** – Print to the terminal the name of the computer in the network.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ hostname
DESKTOP-81VCQE2
```

4. **arch** – Prints the instruction set architecture of the processor. Report on what type of hardware processor (CPU) the computer uses. The

x86_64 label means the computer has a standard 64-bit processor.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ arch
x86_64
```

5. **uname -a** - The command `uname -a` prints detailed system information about the computer. The word `uname` means "Unix name," and the option `-a` means "all," so it displays all available details.

   The output includes:

   `Linux` – the kernel name

   `carmi-VivoBook-ASUSLaptop-TP420UA-TM420UA` – the PC's hostname .

   `6.8.0-106-generic` – the kernel release version.

   `#106~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Fri Mar 6 08:44:59 UTC 2025` – the kernel builds information.

   `x86_64 x86_64 x86_64` – the machine, processor, and hardware platform architecture.

   `GNU/Linux` – the operating system environment.

   "uname" command is useful for checking the system's kernel version, hostname, and architecture.

```
carmi@carmi-VivoBook-ASUSLaptop-TP420UA-TM420UA:~$ uname -a
Linux carmi-VivoBook-ASUSLaptop-TP420UA-TM420UA 6.8.0-106-generic #106~22.04.1-Ubuntu
 SMP PREEMPT_DYNAMIC Fri Mar  6 08:44:59 UTC  x86_64 x86_64 x86_64 GNU/Linux
```

6. **uptime** - Shows how long the Linux system has been running since the last reboot and provides data on the number of tasks waiting for the CPU to complete.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ uptime
 19:31:10 up 3 min,  1 user,  load average: 0.00, 0.00, 0.00
```

7. **Whoami** - Prints the name of the user connected to the terminal.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ whoami
carmifr
```

8. **Who** - Gives a list of all people connected to the machine. It gives information regarding the user connected, the window\ process they are using, and the time they started their session.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ who
carmifr  pts/1        2026-03-28 19:27
```

9. **Id** - This command says who the user is on the system and to which groups the user belongs:
   “uid” - user id is 1000, and username is "carmi".
   “gid” - primary group ID is 1000, username "carmi".
   “groups” - These are all the groups your user belongs to, below are some   examples:
      "adm" - lets the user read some system logs.
      "cdrom" - access to CD/DVD devices.
      "sudo" - means your user can run admin commands with "sudo".
      "plugdev" - access to removable/plugged-in devices like USB hardware.
      “users” - normal user group.
      “docker” - Run Docker commands without "sudo".

```
carmi@carmi-VivoBook-ASUSLaptop-TP420UA-TM420UA:~$ id
uid=1000(carmi) gid=1000(carmi) groups=1000(carmi),4(adm),20(dialout),24(cdrom),27(sudo)
,30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare),138(docker),139(wireshark)
```

10. **Last -** Displays a historical list of user logins, logouts, and system reboots. It retrieves this data from the system’s log file.
    <u>Output Breakdown (Left to right):</u>
    **User:** The username of the account (tomerben) or system events (reboot).
    **Terminal/Line:** The interface used for the session (tty2 for a local text console, seat0 for the local physical login screen, or system boot).
    **Login Time:** The exact day, date, and time the session started.
    **Status/Duration:** Indicates how the session ended and its total length:
    A. <u>Still logged in/still running:</u> The user is currently active, or the system is currently up.
    B. <u>Crash:</u> Indicates the session ended abruptly (via a forced shutdown, power loss, or UI crash). The total time spent in that session is shown in parentheses.

```
tomerbenb13@ubuntu:~$ last
tomerben tty2         tty2             Mon Mar 30 07:11   still logged in
tomerben seat0        login screen     Mon Mar 30 07:11   still logged in
reboot   system boot  6.17.0-14-generi Mon Mar 30 07:11   still running
reboot   system boot  6.17.0-14-generi Mon Mar 30 07:00   still running
tomerben tty2         tty2             Thu Mar 26 21:03 - crash (3+09:57)
tomerben seat0        login screen     Thu Mar 26 21:03 - crash (3+09:57)
reboot   system boot  6.17.0-14-generi Thu Mar 26 21:02   still running
tomerben tty2         tty2             Wed Mar 11 15:56 - crash (15+05:06)
tomerben seat0        login screen     Wed Mar 11 15:56 - crash (15+05:06)
reboot   system boot  6.17.0-14-generi Wed Mar 11 15:55   still running
tomerben tty2         tty2             Wed Mar 11 15:47 - crash  (00:07)
tomerben seat0        login screen     Wed Mar 11 15:47 - crash  (00:07)
reboot   system boot  6.17.0-14-generi Wed Mar 11 15:46   still running

wtmp begins Wed Mar 11 15:46:59 2026
```

11. **w** - This command shows who is currently logged in to the system and what. Shows for each logged-in user:
**USER** – the username.
**TTY** – the terminal for the session.
**FROM** – the remote host (or - if none).
**LOGIN@** – the time the user logged in.
**IDLE** – how long the user has not typed or run anything in the terminal.
**JCPU** – total CPU time for all processes attached to the terminal.
**PCPU** – CPU time for the current process.
**WHAT** – the command currently being run (e.g., bash).

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ w
 13:02:57 up 47 min,  1 user,  load average: 0.00, 0.00, 0.00
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
carmifr  pts/1    -                12:18    44:55  0.01s  0.01s -bash
```

12. **Top -** Provides a dynamic, real-time view of a running system. It displays system summary information alongside a continually updated list of processes currently being managed by the kernel.
<u>Output Breakdown:</u>

**System Summary (Top Section):**
**Row 1 (System Status):** Shows current time, system uptime, number of logged-in users, and system load averages (over 1, 5, and 15 minutes).
**Tasks:** Total number of processes and their current execution states (running, sleeping, stopped, or zombie).
**%Cpu(s):** Real-time CPU usage statistics. Key metrics include - us (time running user processes), - sy (time running system processes), and - id (time idle). Here it shows that the system is mostly idle (99.1 id).
**MiB Mem / Swap:** Displays physical RAM and Swap space utilization, broken down by total, free, used, and buffered/cached amounts.

**Process List (Bottom Section):**
**PID:** Process ID, a unique identifier assigned to each running task.
**USER:** The username of the account that started the process.
**PR & NI:** Priority and Nice values, which determine the scheduling priority of the process.
**VIRT, RES, SHR:** Various metrics of memory usage (Virtual memory, Resident/physical memory, and Shared memory).
**%CPU & %MEM:** The percentage of the total CPU time and physical memory the process is currently consuming.
**COMMAND:** The name of the program or command running the task.

```
tomerbenb13@ubuntu:~$ top

top - 08:42:27 up 3 min,  1 user,  load average: 0.19, 0.37, 0.17
Tasks: 227 total,   1 running, 226 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.4 us,  0.2 sy,  0.0 ni, 99.1 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   8786.4 total,   6393.0 free,   1150.0 used,   1518.1 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   7636.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2936 tomerbe+  20   0 4501164 411772 149332 S   1.6   4.6   0:11.78 gnome-s+
   3878 tomerbe+  20   0  562760  54340  43512 S   0.3   0.6   0:01.22 gnome-t+
      1 root      20   0   23136  14428   9792 S   0.0   0.2   0:02.12 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker+
     10 root      20   0       0      0      0 I   0.0   0.0   0:00.14 kworker+
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.07 ksoftir+
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.13 rcu_pre+
```

13. **echo {con,pre}{sent,fer}{s,ed}** - Bash performs brace expansion, generating all possible combinations by taking one string from each brace group, concatenating them, and then prints (echo) the results.

```
carmifr@DESKTOP-81VCQE2:/mnt/c/Users$ echo {con,pre}{sent,fer}{s,ed}
consents consented confers confered presents presented prefers prefered
```

14. **man ls -** Opens the system manual (documentation) page for the ‘ls’ command. The ‘man’ utility acts as the built-in reference manual, providing detailed instructions, options, and expected behaviors for UNIX commands and programs.

```
tomerbenb13@ubuntu:~$ man ls
```

<u>Output Breakdown:</u>
**NAME:** Displays the command's name alongside a brief description
**SYNOPSIS:** Shows the formal syntax, illustrating how to properly structure the command with its various optional flags and arguments.
**DESCRIPTION:** Provides a comprehensive, in-depth explanation of the command's functionality. It includes a detailed list of all available options (such as ‘ -a ’ to show hidden files or ‘-l ’ for the long listing format) and explains exactly what each one does.
**Navigation/Exit:** The manual opens in a pager interface, allowing you to scroll through the text using the arrow keys. To quit the manual and return to the standard terminal prompt, pressing ‘q’ (for quitting) is necessary.

**Screenshots are down below

```
LS(1)                          User Commands                          LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information  about  the FILEs (the current directory by default).
       Sort entries alphabetically if none of -cftuvSUX nor --sort  is  speci-
       fied.

       Mandatory  arguments  to  long  options are mandatory for short options
       too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters

       --block-size=SIZE
              with  -l,  scale  sizes  by  SIZE  when  printing  them;   e.g.,
              '--block-size=M'; see SIZE format below

       -B, --ignore-backups
              do not list implied entries ending with ~

       -c     with  -lt: sort by, and show, ctime (time of last change of file
              status information); with -l: show ctime and sort by name;  oth-
              erwise: sort by ctime, newest first
```

```
--color[=WHEN]
       color the output WHEN; more info below

-d, --directory
       list directories themselves, not their contents

-D, --dired
       generate output designed for Emacs' dired mode

-f     list all entries in directory order

-F, --classify[=WHEN]
       append indicator (one of */=>@|) to entries WHEN

--file-type
       likewise, except do not append '*'
```

```
--format=WORD
       across  -x, commas -m, horizontal -x, long -l, single-column -1,
       verbose -l, vertical -C

--full-time
       like -l --time-style=full-iso

-g     like -l, but do not list owner

--group-directories-first
       group directories before files; can be augmented with  a  --sort
       option, but any use of --sort=none (-U) disables grouping

-G, --no-group
       in a long listing, don't print group names

-h, --human-readable
       with -l and -s, print sizes like 1K 234M 2G etc.

--si   likewise, but use powers of 1000 not 1024
```

```
-H, --dereference-command-line
       follow symbolic links listed on the command line

--dereference-command-line-symlink-to-dir
       follow  each  command line symbolic link that points to a direc-
       tory

--hide=PATTERN
       do not list implied entries matching shell  PATTERN  (overridden
       by -a or -A)

--hyperlink[=WHEN]
       hyperlink file names WHEN

--indicator-style=WORD
       append indicator with style WORD to entry names: none (default),
       slash (-p), file-type (--file-type), classify (-F)

-i, --inode
       print the index number of each file

-I, --ignore=PATTERN
       do not list implied entries matching shell PATTERN

-k, --kibibytes
       default  to  1024-byte  blocks  for file system usage; used only
       with -s and per directory totals

-l     use a long listing format

-L, --dereference
       when showing file information for a symbolic link, show informa-
       tion for the file the link references rather than for  the  link
       itself
```

**-m** fill width with a comma separated list of entries

**-n**, **--numeric-uid-gid**
like **-l**, but list numeric user and group IDs

**-N**, **--literal**
print entry names without quoting

**-o** like **-l**, but do not list group information

**-p**, **--indicator-style**=slash
append / indicator to directories

**-q**, **--hide-control-chars**
print ? instead of nongraphic characters

**--show-control-chars**
show nongraphic characters as-is (the default, unless program is 'ls' and output is a terminal)

**-Q**, **--quote-name**
enclose entry names in double quotes

**--quoting-style**=WORD
use quoting style WORD for entry names: literal, locale, shell, shell-always, shell-escape, shell-escape-always, c, escape (overrides QUOTING_STYLE environment variable)

**-r**, **--reverse**
reverse order while sorting

**-R**, **--recursive**
list subdirectories recursively

**-s**, **--size**
print the allocated size of each file, in blocks

**-S** sort by file size, largest first

**--sort**=WORD
sort by WORD instead of name: none (**-U**), size (**-S**), time (**-t**), version (**-v**), extension (**-X**), width

**--time**=WORD
select which timestamp used to display or sort; access time (**-u**): atime, access, use; metadata change time (**-c**): ctime, status; modified time (default): mtime, modification; birth time: birth, creation;

with **-l**, WORD determines which time to show; with **--sort**=time, sort by WORD (newest first)

**--time-style**=TIME_STYLE
time/date format with **-l**; see TIME_STYLE below

```
-t     sort by time, newest first; see --time

-T, --tabsize=COLS
       assume tab stops at each COLS instead of 8

-u     with  -lt:  sort by, and show, access time; with -l: show access
       time and sort by name; otherwise: sort by  access  time,  newest
       first

-U     do not sort; list entries in directory order

-v     natural sort of (version) numbers within text

-w, --width=COLS
       set output width to COLS.  0 means no limit

-x     list entries by lines instead of by columns

-X     sort alphabetically by entry extension

-Z, --context
       print any security context of each file

--zero end each output line with NUL, not newline

-1     list one file per line

--help display this help and exit

--version
       output version information and exit

The  SIZE  argument  is  an  integer and optional unit (example: 10K is
10*1024).  Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or  KB,MB,...
(powers  of 1000).  Binary prefixes can be used, too: KiB=K, MiB=M, and
so on.

The TIME_STYLE argument can be  full-iso,  long-iso,  iso,  locale,  or
+FORMAT.   FORMAT  is  interpreted  like in date(1).  If FORMAT is FOR-
MAT1<newline>FORMAT2, then FORMAT1 applies to non-recent files and FOR-
MAT2 to recent files.  TIME_STYLE prefixed with 'posix-'  takes  effect
only  outside  the POSIX locale.  Also the TIME_STYLE environment vari-
able sets the default style to use.

The WHEN argument defaults to  'always'  and  can  also  be  'auto'  or
'never'.

Using  color  to distinguish file types is disabled both by default and
with --color=never.  With --color=auto, ls emits color codes only  when
standard  output is connected to a terminal.  The LS_COLORS environment
variable can change the settings.  Use the dircolors(1) command to  set
it.
```

```
  Exit status:
      0      if OK,

      1      if minor problems (e.g., cannot access subdirectory),

      2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR
      Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
      GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
      Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
      Copyright © 2023 Free Software Foundation, Inc.  License GPLv3+: GNU
      GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
      This is free software: you are free  to  change  and  redistribute  it.
      There is NO WARRANTY, to the extent permitted by law.

SEE ALSO
      dircolors(1)

      Full documentation <https://www.gnu.org/software/coreutils/ls>
      or available locally via: info '(coreutils) ls invocation'

GNU coreutils 9.4                 June 2025                          LS(1)
 Manual page ls(1) line 211/249 (END) (press h for help or q to quit)
```

15. **man who** – "man" (short for manual) command opens the documentation page of its following command, which is, in this case, the "who" command.

```
WHO(1)                           User Commands                          WHO(1)

NAME
       who - show who is logged on

SYNOPSIS
       who [OPTION]... [ FILE | ARG1 ARG2 ]

DESCRIPTION
       Print information about users who are currently logged in.

       -a, --all
              same as -b -d --login -p -r -t -T -u

       -b, --boot
              time of last system boot

       -d, --dead
              print dead processes

       -H, --heading
              print line of column headings

       -l, --login
              print system login processes

       --lookup
              attempt to canonicalize hostnames via DNS

       -m     only hostname and user associated with stdin
```

```
       -p, --process
              print active processes spawned by init

       -q, --count
              all login names and number of users logged on

       -r, --runlevel
              print current runlevel

       -s, --short
              print only name, line, and time (default)

       -t, --time
              print last system clock change

       -T, -w, --mesg
              add user's message status as +, - or ?

       -u, --users
 Manual page who(1) line 1 (press h for help or q to quit)
```

16. **Clear -** Clears the terminal screen to provide a clean visual workspace.

```
tomerbenb13@ubuntu:~$ echo hello world
hello world
tomerbenb13@ubuntu:~$ w
 07:34:20 up 39 min,  1 user,  load average: 0.01, 0.03, 0.02
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
tomerben tty2     -                06:55   39:16   0.03s  0.03s /usr/libexec/gnome-ses
tomerbenb13@ubuntu:~$ date
Tue Mar 31 07:34:39 AM UTC 2026
tomerbenb13@ubuntu:~$ hostname
ubuntu
tomerbenb13@ubuntu:~$ whoami
tomerbenb13
tomerbenb13@ubuntu:~$ clear
```

```
tomerbenb13@ubuntu:~$
```

17. **bc -l -** Starts ‘bc’ (Basic Calculator), which is an arbitrary-precision calculator language, in an interactive mode. The ‘-l’ flag specifically tells the program to load the standard math library. This enables advanced mathematical functions (like sine, cosine, and logarithms) and changes the default decimal precision (scale) from 0 to 20 digits.

Output Breakdown:
**Introductory Banner:** The text displayed in your screenshot shows the program's version (1.07.1), copyright details, and a standard open-source warranty disclaimer.
**Interactive Mode:** The calculator is now running and waiting for an input of mathematical expressions.
**Exit:** to exit this calculator environment and return to the normal UNIX shell, typing ‘quit’ + ‘Enter’ is needed.

```
tomerbenb13@ubuntu:~$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Founda
tion, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
5+4
9
12*12
144
18/2
9.00000000000000000000
2^3
8
quit
```

18. **echo 5+4 | bc -l -** Evaluates a mathematical expression directly from the command line and returns the result without keeping the interactive calculator open.
    **Echo 5+4:** The 'echo' command prints the literal string "5+4" to the standard output.
    **| (Pipe):** The pipe operator is a fundamental UNIX concept. It captures the output from the command on its left (the string "5+4") and feeds it directly as the standard input into the command on its right (bc -l).

    Because the input was provided directly through the pipe rather than typed interactively, 'bc' processes the calculation, displays the answer, and immediately terminates, returning to the normal terminal prompt.

```
tomerbenb13@ubuntu:~$ echo 5+4 | bc -l
9
```

19. **yes please -** Repeatedly outputs the provided string ("please") to the terminal until it is forcibly stopped.

    **Infinite Loop:** The 'yes' utility takes the text argument provided ("please") and prints it continuously, line after line, without pausing. If no argument is provided, it defaults to printing "y".
    **Standard Use Case:** While it seems like a chaotic loop here, 'yes' is traditionally used in automated scripts. It is used to pipe a confirmation string (usually "y") into other commands that require interactive user prompts, such as confirming multiple file deletions or software installations.
    **Exit:** Because this command runs infinitely and takes over the standard output, we must manually interrupt the process by pressing 'Ctrl + c'.

```
please
please
please
please
^Cease
tomerbenb13@ubuntu:~$
```

20. **time sleep 5 -** Pauses the terminal for exactly 5 seconds and measures the exact amount of time it took for that process to complete.

    After the 5-second pause, the terminal will display three specific metrics:
    **real:** The actual time that passed from start to finish (which will be slightly over 5.000 seconds).
    **user:** The amount of CPU time spent executing user-space code. This will be approximately 0.000s, as "sleeping" does not actively consume CPU processing cycles.
    **sys:** The amount of CPU time spent by the system kernel, which will also be approximately 0.000s (here it is 0.003s).

```
tomerbenb13@ubuntu:~$ time sleep 5

real    0m5.015s
user    0m0.000s
sys     0m0.003s
```

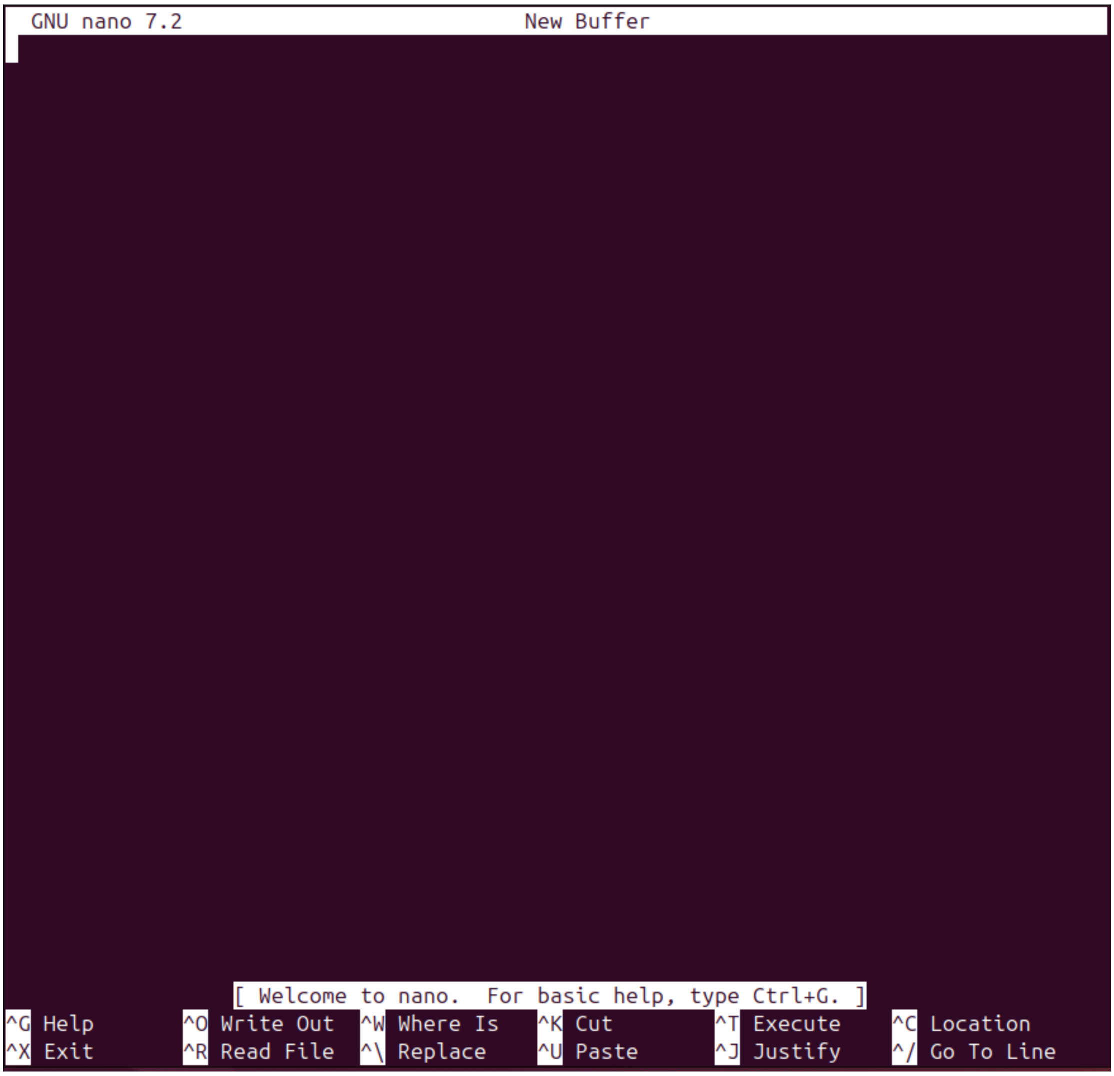
21. **History -** Displays a numbered list of the commands previously executed in the user's terminal sessions.

```
tomerbenb13@ubuntu:~$ history
    1  sudo apt-get install bzip2 tar
    2  man ls
    3  clear
    4  echo hello world
    5  w
    6  date
    7  hostname
    8  whoami
    9  clear
   10  bc -l
   11  echo 5+4 | bc -l
   12  yes please
   13  time sleep 5
   14  history
```

22. **Nano -** Opens the Nano text editor, which is a command-line text editor used to create and modify text files directly within the terminal.
    When typing ‘nano filename.txt’, it opens that specific file, and typing just ‘nano’, it opens a temporary blank buffer.

**Typing and Editing:** We can immediately start typing to add text. We can use the arrow keys to navigate through the document, making it much more intuitive for beginners.

**The Shortcut Menu:** At the bottom of the screen, ‘nano’ displays a very helpful menu of keyboard shortcuts. The caret symbol (^) represents the ‘Ctrl’ key.



# Part 2 Unix File system, Links, and quotes

1. We tried the next command sequence and got:

   1. **Try the following command sequence[ copy paste the output into your report]:**
      - **cd**
      - **pwd**
      - **ls -al**
      - **cd .**
      - **pwd**
      - **cd ..**
      - **pwd**
      - **ls -al**
      - **cd ..**
      - **pwd**
      - **ls -al**
      - **cd ..**
      - **pwd**
      - **cd /etc**
      - **ls -al | more**
      - **cat passwd**
      - **cd -**
      - **pwd**

```
tomerbenb13@ubuntu:~$ cd
tomerbenb13@ubuntu:~$ pwd
/home/tomerbenb13
tomerbenb13@ubuntu:~$ ls -al
total 112
drwxr-x--- 16 tomerbenb13 tomerbenb13 4096 Apr  5 07:39 .
drwxr-xr-x  3 root        root        4096 Mar 11 15:47 ..
-rw-------  1 tomerbenb13 tomerbenb13  398 Mar 31 09:59 .bash_history
-rw-r--r--  1 tomerbenb13 tomerbenb13  220 Mar 31  2024 .bash_logout
-rw-r--r--  1 tomerbenb13 tomerbenb13 3771 Mar 31  2024 .bashrc
drwx------ 10 tomerbenb13 tomerbenb13 4096 Mar 11 15:53 .cache
drwx------ 12 tomerbenb13 tomerbenb13 4096 Mar 26 21:20 .config
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Desktop
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Documents
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Downloads
drwx------  2 tomerbenb13 tomerbenb13 4096 Mar 30 08:40 .gnupg
-rw-------  1 tomerbenb13 tomerbenb13   20 Mar 31 07:27 .lesshst
drwx------  4 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 .local
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Music
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Pictures
-rw-r--r--  1 tomerbenb13 tomerbenb13  807 Mar 31  2024 .profile
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Public
drwx------  4 tomerbenb13 tomerbenb13 4096 Mar 30 09:00 snap
drwx------  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 .ssh
-rw-r--r--  1 tomerbenb13 tomerbenb13    0 Mar 11 15:52 .sudo_as_admin_successful
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Templates
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-clipboard-tty2-control.pid
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-clipboard-tty2-service.pid
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-draganddrop-tty2-control.pid
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-hostversion-tty2-control.pid
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-seamless-tty2-control.pid
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-vmsvga-session-tty2-control.pid
-rw-r-----  1 tomerbenb13 tomerbenb13    5 Apr  5 07:39 .vboxclient-vmsvga-session-tty2-service.pid
drwxr-xr-x  2 tomerbenb13 tomerbenb13 4096 Mar 11 15:47 Videos
tomerbenb13@ubuntu:~$ cd .
tomerbenb13@ubuntu:~$ pwd
/home/tomerbenb13
```

```
tomerbenb13@ubuntu:~$ cd ..
tomerbenb13@ubuntu:/home$ pwd
/home
tomerbenb13@ubuntu:/home$ ls -al
total 12
drwxr-xr-x  3 root        root        4096 Mar 11 15:47 .
drwxr-xr-x 23 root        root        4096 Mar 11 15:39 ..
drwxr-x--- 16 tomerbenb13 tomerbenb13 4096 Apr  5 07:39 tomerbenb13
tomerbenb13@ubuntu:/home$ cd ..
tomerbenb13@ubuntu:/$ pwd
/
tomerbenb13@ubuntu:/$ ls -al
total 96
drwxr-xr-x  23 root root  4096 Mar 11 15:39 .
drwxr-xr-x  23 root root  4096 Mar 11 15:39 ..
lrwxrwxrwx   1 root root     7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root  4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   3 root root  4096 Mar 11 15:44 boot
dr-xr-xr-x   2 root root  4096 Mar 11 15:37 cdrom
drwxr-xr-x  19 root root  4080 Apr  5 07:38 dev
drwxr-xr-x 138 root root 12288 Mar 11 15:53 etc
drwxr-xr-x   3 root root  4096 Mar 11 15:47 home
lrwxrwxrwx   1 root root     7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx   1 root root     9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x   2 root root  4096 Apr  8  2024 lib.usr-is-merged
drwx------   2 root root 16384 Mar 11 15:39 lost+found
drwxr-xr-x   3 root root  4096 Mar 11 15:50 media
drwxr-xr-x   2 root root  4096 Feb 10 00:19 mnt
drwxr-xr-x   3 root root  4096 Mar 11 15:53 opt
dr-xr-xr-x 275 root root     0 Apr  5 07:38 proc
drwx------   5 root root  4096 Mar 11 15:47 root
drwxr-xr-x  34 root root   880 Apr  5 07:38 run
lrwxrwxrwx   1 root root     8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x   2 root root  4096 Mar 31  2024 sbin.usr-is-merged
drwxr-xr-x  12 root root  4096 Feb 10 00:32 snap
drwxr-xr-x   2 root root  4096 Feb 10 00:19 srv
dr-xr-xr-x  13 root root     0 Apr  5 07:38 sys
drwxrwxrwt  16 root root  4096 Apr  5 07:48 tmp
```

```
drwxr-xr-x  12 root root  4096 Feb 10 00:19 usr
drwxr-xr-x  14 root root  4096 Mar 11 15:47 var
tomerbenb13@ubuntu:/$ cd ..
tomerbenb13@ubuntu:/$ pwd
/
tomerbenb13@ubuntu:/$ cd /etc
tomerbenb13@ubuntu:/etc$ ls -al | more
total 1144
drwxr-xr-x 138 root                 root                 12288 Mar 11 15:53 .
drwxr-xr-x  23 root                 root                  4096 Mar 11 15:39 ..
-rw-r--r--   1 root                 root                  3444 Jul  5  2023 addu
ser.conf
drwxr-xr-x   3 root                 root                  4096 Feb 10 00:28 alsa
drwxr-xr-x   2 root                 root                  4096 Feb 10 00:31 alte
rnatives
-rw-r--r--   1 root                 root                   335 Apr  8  2024 anac
rontab
-rw-r--r--   1 root                 root                   433 Apr  8  2024 apg.
conf
drwxr-xr-x   5 root                 root                  4096 Feb 10 00:27 apm
drwxr-xr-x   2 root                 root                  4096 Feb 10 00:27 appa
rmor
drwxr-xr-x   9 root                 root                  4096 Feb 10 00:31 appa
rmor.d
drwxr-xr-x   3 root                 root                  4096 Feb 10 00:29 appo
rt
drwxr-xr-x   9 root                 root                  4096 Mar 11 15:39 apt
drwxr-xr-x   3 root                 root                  4096 Feb 10 00:29 avah
i
-rw-r--r--   1 root                 root                  2319 Mar 31  2024 bash
.bashrc
-rw-r--r--   1 root                 root                    45 Jan 24  2020 bash
_completion
```

```
-rw-r--r--   1 root                root                 367 Aug  2  2022 bind
resvport.blacklist
drwxr-xr-x   2 root                root                4096 Apr 19  2024 binf
mt.d
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 blue
```

```
tooth
-rw-r-----   1 root                root                  33 Feb 10 00:31 brla
pi.key
drwxr-xr-x   7 root                root                4096 Feb 10 00:28 brlt
ty
-rw-r--r--   1 root                root               30571 Mar 31  2024 brlt
ty.conf
drwxr-xr-x   3 root                root                4096 Feb 10 00:19 ca-c
ertificates
-rw-r--r--   1 root                root                6288 Feb 10 00:20 ca-c
ertificates.conf
drwxr-s---   2 root                dip                 4096 Feb 10 00:29 chat
scripts
drwxr-xr-x   5 root                root                4096 Mar 11 15:47 clou
d
drwxr-xr-x   2 colord              colord              4096 Mar 11 15:47 colo
rd
drwxr-xr-x   2 root                root                4096 Mar 11 15:42 cons
ole-setup
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 crac
klib
drwx------   2 root                root                4096 Apr 19  2024 cred
store
drwx------   2 root                root                4096 Apr 19  2024 cred
store.encrypted
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 cron
.d
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 cron
.daily
drwxr-xr-x   2 root                root                4096 Feb 10 00:19 cron
.hourly
drwxr-xr-x   2 root                root                4096 Feb 10 00:28 cron
.monthly
-rw-r--r--   1 root                root                1136 Mar 31  2024 cron
tab
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 cron
.weekly
drwxr-xr-x   2 root                root                4096 Feb 10 00:19 cron
```


```
.hourly
drwxr-xr-x   2 root                root                4096 Feb 10 00:28 cron
.monthly
-rw-r--r--   1 root                root                1136 Mar 31  2024 cron
tab
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 cron
.weekly
drwxr-xr-x   2 root                root                4096 Feb 10 00:19 cron
```


```
.yearly
drwxr-xr-x   5 root                lp                  4096 Apr  5 07:39 cups
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 cups
helpers
drwxr-xr-x   4 root                root                4096 Feb 10 00:19 dbus
-1
drwxr-xr-x   4 root                root                4096 Feb 10 00:28 dcon
f
-rw-r--r--   1 root                root                2967 Apr 12  2024 debc
onf.conf
-rw-r--r--   1 root                root                  11 Apr 22  2024 debi
an_version
drwxr-xr-x   2 root                root                4096 Feb 10 00:28 debu
ginfod
drwxr-xr-x   3 root                root                4096 Mar 11 15:44 defa
ult
```

```
-rw-r--r--   1 root                root                1706 Jul  5  2023 delu
ser.conf
drwxr-xr-x   2 root                root                4096 Feb 10 00:21 depm
od.d
drwxr-xr-x   3 root                root                4096 Feb 10 00:20 dhcp
-rw-r--r--   1 root                root                1429 Mar 31  2024 dhcp
cd.conf
drwxr-xr-x   2 root                root                4096 Mar 11 15:45 dict
ionaries-common
drwxr-xr-x   4 root                root                4096 Feb 10 00:21 dpkg
-rw-r--r--   1 root                root                 685 Apr  8  2024 e2sc
rub.conf
drwxr-xr-x   3 root                root                4096 Feb 10 00:27 emac
s
```

```
-rw-r--r--   1 root                root                 106 Feb 10 00:19 envi
ronment
drwxr-xr-x   2 root                root                4096 Feb 10 00:31 envi
ronment.d
-rw-r--r--   1 root                root                1853 Oct 17  2022 ethe
rtypes
drwxr-xr-x   5 root                root                4096 Feb 10 00:29 font
s
-rw-r--r--   1 root                root                  20 Apr  4  2024 fpri
ntd.conf
-rw-r--r--   1 root                root                 446 Mar 11 15:44 fsta
b
-rw-r--r--   1 root                root                 694 Apr  8  2024 fuse
.conf
drwxr-xr-x   4 root                root                4096 Feb 10 00:29 fwup
d
-rw-r--r--   1 root                root                2584 Jan 31  2024 gai.
conf
drwxr-xr-x   3 root                root                4096 Feb 10 00:29 gdb
drwxr-xr-x   8 root                root                4096 Feb 10 00:31 gdm3
drwxr-xr-x   3 root                root                4096 Feb 10 00:31 geoc
lue
drwxr-xr-x   4 root                root                4096 Feb 10 00:27 ghos
tscript
drwxr-xr-x   3 root                root                4096 Feb 10 00:27 glvn
d
drwxr-xr-x   2 root                root                4096 Feb 10 00:28 gnom
e
drwxr-xr-x   2 gnome-remote-desktop gnome-remote-desktop  4096 Feb 10 00:31 gnom
e-remote-desktop
drwxr-xr-x   2 root                root                4096 Mar 11 15:46 gnut
ls
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 grof
f
-rw-r--r--   1 root                root                1115 Mar 11 15:53 grou
p
-rw-r--r--   1 root                root                1097 Mar 11 15:53 grou
p-
```

```
drwxr-xr-x   2 root                root                4096 Feb 10 00:33 grub
.d
-rw-r-----   1 root                shadow               936 Mar 11 15:53 gsha
dow
-rw-r-----   1 root                shadow               921 Mar 11 15:53 gsha
dow-
drwxr-xr-x   3 root                root                4096 Feb 10 00:19 gss
drwxr-xr-x   2 root                root                4096 Feb 10 00:29 gtk-
2.0
drwxr-xr-x   2 root                root                4096 Feb 10 00:31 gtk-
3.0
-rw-r--r--   1 root                root                4436 Oct  6  2022 hdpa
rm.conf
-rw-r--r--   1 root                root                  92 Apr 22  2024 host
.conf
-rw-r--r--   1 root                root                   7 Mar 11 15:47 host
name
```

```
-rw-r--r--   1 root                 root                 273 Feb  4  2025 host
s
-rw-r--r--   1 root                 root                 411 Feb 10 00:28 host
s.allow
-rw-r--r--   1 root                 root                 711 Feb 10 00:28 host
s.deny
drwxr-xr-x   2 root                 root                4096 Feb 10 00:31 hp
drwxr-xr-x   3 root                 root                4096 Feb 10 00:28 ifpl
ugd
drwxr-xr-x   2 root                 root                4096 Feb 10 00:29 init
drwxr-xr-x   2 root                 root                4096 Feb 10 00:31 init
.d
drwxr-xr-x   5 root                 root                4096 Feb 10 00:29 init
ramfs-tools
-rw-r--r--   1 root                 root                1875 Mar 31  2024 inpu
trc
drwxr-xr-x   2 root                 root                4096 Feb 10 00:31 inss
erv.conf.d
drwxr-xr-x   2 root                 root                4096 Feb 10 00:29 ipp-
usb
drwxr-xr-x   4 root                 root                4096 Feb 10 00:21 ipro
```

```
ute2
-rw-r--r--   1 root                 root                  26 Feb  6 07:23 issu
e
-rw-r--r--   1 root                 root                  19 Feb  6 07:23 issu
e.net
drwxr-xr-x   7 root                 root                4096 Mar 11 15:53 kern
el
-rw-r--r--   1 root                 root                1308 Mar 31  2024 kern
eloops.conf
drwxr-xr-x   2 root                 root                4096 Feb 10 00:29 krb5
.conf.d
drwxr-xr-x   2 root                 root                4096 Feb 10 00:28 ldap
-rw-r--r--   1 root                 root               55283 Mar 11 15:53 ld.s
o.cache
-rw-r--r--   1 root                 root                  34 Aug  2  2022 ld.s
o.conf
drwxr-xr-x   2 root                 root                4096 Feb 10 00:21 ld.s
o.conf.d
-rw-r--r--   1 root                 root                 267 Apr 22  2024 lega
l
-rw-r--r--   1 root                 root                  27 Apr  8  2024 liba
o.conf
-rw-r--r--   1 root                 root                 191 Mar 31  2024 liba
udit.conf
drwxr-xr-x   3 root                 root                4096 Feb 10 00:28 libb
lockdev
drwxr-xr-x   2 root                 root                4096 Feb 10 00:29 libi
bverbs.d
drwxr-xr-x   2 root                 root                4096 Feb 10 00:29 libn
l-3
drwxr-xr-x   2 root                 root                4096 Apr  8  2024 libp
aper.d
-rw-r--r--   1 root                 root                2996 Mar 30  2024 loca
le.alias
-rw-r--r--   1 root                 root                  17 Mar 11 15:47 loca
le.conf
-rw-r--r--   1 root                 root                9563 Mar 11 15:47 loca
le.gen
```

```
lrwxrwxrwx   1 root                 root                  27 Mar 11 15:47 loca
ltime -> /usr/share/zoneinfo/Etc/UTC
drwxr-xr-x   4 root                 root                4096 Feb 10 00:27 logc
heck
-rw-r--r--   1 root                 root               12345 Feb 22  2024 logi
n.defs
-rw-r--r--   1 root                 root                 586 Apr  8  2024 logr
otate.conf
drwxr-xr-x   2 root                 root                4096 Feb 10 00:31 logr
otate.d
```

```
-rw-r--r--   1 root                root                  104 Feb  6 07:22 lsb-
release
-r--r--r--   1 root                root                   33 Mar 11 15:42 mach
ine-id
-rw-r--r--   1 root                root                  111 Mar 31  2024 magi
c
-rw-r--r--   1 root                root                  111 Mar 31  2024 magi
c.mime
-rw-r--r--   1 root                root                 5230 Apr  8  2024 manp
ath.config
-rw-r--r--   1 root                root                75113 Jul 12  2023 mime
.types
-rw-r--r--   1 root                root                  744 Apr  8  2024 mke2
fs.conf
drwxr-xr-x   4 root                root                 4096 Feb 10 00:28 Mode
mManager
drwxr-xr-x   2 root                root                 4096 Mar 11 15:44 modp
robe.d
-rw-r--r--   1 root                root                  212 Feb 10 00:20 modu
les
drwxr-xr-x   2 root                root                 4096 Feb 10 00:29 modu
les-load.d
lrwxrwxrwx   1 root                root                   19 Feb 10 00:19 mtab
 -> ../proc/self/mounts
-rw-r--r--   1 root                root                11424 May 23  2023 nano
rc
-rw-r--r--   1 root                root                  767 Mar 31  2024 netc
onfig
```

```
drwxr-xr-x   2 root                root                 4096 Mar 11 15:47 netp
lan
drwxr-xr-x   6 root                root                 4096 Feb 10 00:28 netw
ork
drwxr-xr-x   8 root                root                 4096 Feb 10 00:20 netw
orkd-dispatcher
drwxr-xr-x   8 root                root                 4096 Feb 10 00:31 Netw
orkManager
-rw-r--r--   1 root                root                   91 Apr 22  2024 netw
orks
drwxr-xr-x   2 root                root                 4096 Feb 10 00:20 newt
-rwxr-xr-x   1 root                root                  243 Oct 19  2023 nfta
bles.conf
-rw-r--r--   1 root                root                  594 Feb 10 00:29 nssw
itch.conf
drwxr-xr-x   4 root                root                 4096 Feb 10 00:29 open
vpn
drwxr-xr-x   2 root                root                 4096 Feb 10 00:19 opt
lrwxrwxrwx   1 root                root                   21 Feb  6 07:23 os-r
elease -> ../usr/lib/os-release
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 Pack
ageKit
-rw-r--r--   1 root                root                  552 Oct 13  2022 pam.
conf
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 pam.
d
-rw-r--r--   1 root                root                    3 Feb 10 00:28 pape
rsize
-rw-r--r--   1 root                root                 2924 Mar 11 15:53 pass
wd
-rw-r--r--   1 root                root                 2879 Mar 11 15:47 pass
wd-
drwxr-xr-x   2 root                root                 4096 Feb 10 00:29 pcmc
ia
drwxr-xr-x   3 root                root                 4096 Feb 10 00:27 perl
drwxr-xr-x   4 root                root                 4096 Feb 10 00:28 pki
drwxr-xr-x   2 root                root                 4096 Feb 25  2025 plym
outh
```

```
drwxr-xr-x   3 root                root                 4096 Feb 10 00:28 pm
-rw-r--r--   1 root                root                 7649 Feb 10 00:29 pnm2
ppa.conf
drwxr-xr-x   4 root                root                 4096 Feb 10 00:28 polk
it-1
drwxr-xr-x   8 root                dip                  4096 Feb 10 00:29 ppp
-rw-r--r--   1 root                root                  582 Apr 22  2024 prof
ile
drwxr-xr-x   2 root                root                 4096 Feb 10 00:29 prof
ile.d
-rw-r--r--   1 root                root                 3144 Oct 17  2022 prot
ocols
drwxr-xr-x   3 root                root                 4096 Feb 10 00:29 puls
e
-rw-------   1 root                root                    0 Feb 10 00:19 .pwd
.lock
drwxr-xr-x   2 root                root                 4096 Feb 10 00:20 pyth
on3
drwxr-xr-x   2 root                root                 4096 Mar 11 15:46 pyth
on3.12
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc0.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc1.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc2.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc3.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc4.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc5.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 rc6.
d
drwxr-xr-x   2 root                root                 4096 Feb 10 00:29 rcS.
d
lrwxrwxrwx   1 root                root                   39 Feb 10 00:20 reso
```

```
lv.conf -> ../run/systemd/resolve/stub-resolv.conf
-rw-r--r--   1 root                root                  862 Feb 10 00:19 .res
olv.conf.systemd-resolved.bak
lrwxrwxrwx   1 root                root                   13 Apr  8  2024 rmt
-> /usr/sbin/rmt
-rw-r--r--   1 root                root                  911 Oct 17  2022 rpc
-rw-r--r--   1 root                root                 1213 Mar 22  2024 rsys
log.conf
drwxr-xr-x   2 root                root                 4096 Feb 10 00:29 rsys
log.d
-rw-r--r--   1 root                root                 5772 Jan  6  2024 ryge
l.conf
drwxr-xr-x   3 root                root                 4096 Feb 10 00:31 sane
.d
drwxr-xr-x   4 root                root                 4096 Feb 10 00:28 secu
rity
drwxr-xr-x   2 root                root                 4096 Feb 10 00:19 seli
nux
-rw-r--r--   1 root                root                10593 Mar 31  2024 sens
ors3.conf
drwxr-xr-x   2 root                root                 4096 Feb 10 00:28 sens
ors.d
-rw-r--r--   1 root                root                12813 Mar 27  2021 serv
ices
drwxr-xr-x   3 root                root                 4096 Feb 10 00:31 sgml
-rw-r-----   1 root                shadow               1366 Mar 11 15:53 shad
ow
-rw-r-----   1 root                shadow               1344 Mar 11 15:47 shad
ow-
-rw-r--r--   1 root                root                  118 Feb 10 00:19 shel
ls
drwxr-xr-x   2 root                root                 4096 Feb 10 00:19 skel
drwxr-xr-x   2 root                root                 4096 Feb 10 00:28 snmp
drwxr-xr-x   4 root                root                 4096 Feb 10 00:29 spee
```

```
ch-dispatcher
drwxr-xr-x   3 root             root             4096 Mar 11 15:47 ssh
drwxr-xr-x   4 root             root             4096 Feb 10 00:21 ssl
drwx--x--x   3 root             root             4096 Feb 10 00:28 sssd
-rw-r--r--   1 root             root               25 Mar 11 15:47 subg
id
-rw-r--r--   1 root             root                0 Feb 10 00:19 subg
id-
-rw-r--r--   1 root             root               25 Mar 11 15:47 subu
id
-rw-r--r--   1 root             root                0 Feb 10 00:19 subu
id-
-rw-r--r--   1 root             root             4343 Apr  8  2024 sudo
.conf
-r--r-----   1 root             root             1800 Jan 29  2024 sudo
ers
drwxr-xr-x   2 root             root             4096 Feb 10 00:21 sudo
ers.d
-rw-r--r--   1 root             root             9804 Apr  8  2024 sudo
_logsrvd.conf
drwxr-xr-x   2 root             root             4096 Feb 10 00:20 supe
rcat
-rw-r--r--   1 root             root             2209 Mar 24  2024 sysc
tl.conf
drwxr-xr-x   2 root             root             4096 Feb 10 00:21 sysc
tl.d
drwxr-xr-x   2 root             root             4096 Feb 10 00:29 syss
tat
drwxr-xr-x   7 root             root             4096 Mar 11 15:47 syst
emd
drwxr-xr-x   2 root             root             4096 Feb 10 00:19 term
info
drwxr-xr-x   2 root             root             4096 Mar 11 15:44 ther
mald
-rw-r--r--   1 root             root                8 Mar 11 15:47 time
zone
drwxr-xr-x   2 root             root             4096 Apr 19  2024 tmpf
iles.d
```

```
drwxr-xr-x   2 root             root             4096 Feb 10 00:21 ubun
tu-advantage
-rw-r--r--   1 root             root             1260 Jan 27  2023 ucf.
conf
drwxr-xr-x   4 root             root             4096 Feb 10 00:21 udev
drwxr-xr-x   2 root             root             4096 Feb 10 00:29 udis
ks2
drwxr-xr-x   3 root             root             4096 Feb 10 00:29 ufw
-rw-r--r--   1 root             root              208 Feb 10 00:19 .upd
ated
drwxr-xr-x   3 root             root             4096 Feb 10 00:30 upda
te-manager
drwxr-xr-x   2 root             root             4096 Feb 10 00:31 upda
te-motd.d
drwxr-xr-x   2 root             root             4096 Apr  2  2025 upda
te-notifier
drwxr-xr-x   2 root             root             4096 Feb 10 00:29 UPow
er
-rw-r--r--   1 root             root             1523 Apr  8  2024 usb_
modeswitch.conf
drwxr-xr-x   2 root             root             4096 Dec 16  2023 usb_
modeswitch.d
lrwxrwxrwx   1 root             root               16 Feb 10 00:19 vcon
sole.conf -> default/keyboard
drwxr-xr-x   2 root             root             4096 Feb 10 00:21 vim
lrwxrwxrwx   1 root             root               23 Feb 26  2024 vtrg
b -> /etc/alternatives/vtrgb
drwxr-xr-x   5 root             root             4096 Feb 10 00:27 vulk
an
-rw-r--r--   1 root             root             4942 Jun 19  2024 wget
rc
```

```
drwxr-xr-x   2 root                root                 4096 Feb 10 00:29 wpa_
supplicant
drwxr-xr-x  12 root                root                 4096 Feb 10 00:31 X11
-rw-r--r--   1 root                root                  681 Apr  8  2024 xatt
r.conf
drwxr-xr-x   6 root                root                 4096 Feb 10 00:28 xdg
drwxr-xr-x   2 root                root                 4096 Feb 10 00:31 xml
-rw-r--r--   1 root                root                  460 Jan 20  2023 zsh_
```

```
tomerbenb13@ubuntu:/etc$ cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
```

```
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup/:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
tomerbenb13:x:1000:1000:tomerbenb13:/home/tomerbenb13:/bin/bash
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
tomerbenb13@ubuntu:/etc$ cd -
/
tomerbenb13@ubuntu:/$ pwd
/
```

2.

What is the role of the command **cd -** ? prove your claim by writing a few commands and their output

"cd –" - The command cd refers to the current directory, and cd – changes the directory back to the previous directory you were in. Below is proof from the terminal:

```
carmifr@DESKTOP-81VCQE2:/home$ cd -
/
carmifr@DESKTOP-81VCQE2:/$ cd -
/home
carmifr@DESKTOP-81VCQE2:/home$ cd -
/
carmifr@DESKTOP-81VCQE2:/$ cd -
/home
carmifr@DESKTOP-81VCQE2:/home$ cd -
/
carmifr@DESKTOP-81VCQE2:/$
```

3.

3. Explore `/dev`. Can you identify what devices are available? Which are character-oriented and which are block-oriented? Can you identify your tty (terminal) device (typing `who` might help); who is the owner of your tty (use `ls -l`)? *[ write the commands you used to find it out into your report]*

To identify which devices are currently available on your machine, we list the contents of this directory. Every entry generated by the ‘ls -l /dev’ command, such as cpu, disk, tty, console, or loop, represents a device that the operating system currently recognizes and has made available for use.

By looking at the very first letter on the far left of each line in your screenshot, you can easily identify the device types:

- **Character-oriented devices (c):** Look at the entries for ‘autofs’, ‘console’, ‘fb0’, ‘kmsg’. Their lines start with the letter ‘**c**’, meaning they handle data character-by-character.
- **Block-oriented devices (b):** Look at the very bottom of your screenshot. The entries for ‘loop0’, ‘loop1’, etc., all begin with the letter ‘**b**’, indicating they handle data in larger blocks.

We can see that in the screenshot below:

```
tomerbenb13@ubuntu:/$ cd /dev
tomerbenb13@ubuntu:/dev$ ls -l
total 0
crw-r--r--   1 root        root      10, 235 Apr  5 07:38 autofs
drwxr-xr-x   2 root        root          320 Apr  5 07:39 block
drwxr-xr-x   2 root        root           80 Apr  5 07:38 bsg
crw-------   1 root        root      10, 234 Apr  5 07:38 btrfs-control
drwxr-xr-x   3 root        root           60 Apr  5 07:38 bus
lrwxrwxrwx   1 root        root            3 Apr  5 07:38 cdrom -> sr0
drwxr-xr-x   2 root        root         3740 Apr  5 09:05 char
crw-------   1 root        root       5,   1 Apr  5 07:38 console
lrwxrwxrwx   1 root        root           11 Apr  5 07:38 core -> /proc/kcore
drwxr-xr-x   5 root        root          100 Apr  5 07:38 cpu
crw-------   1 root        root      10, 260 Apr  5 07:38 cpu_dma_latency
crw-------   1 root        root      10, 203 Apr  5 07:38 cuse
drwxr-xr-x  10 root        root          200 Apr  5 07:38 disk
drwxr-xr-x   2 root        root           60 Apr  5 07:38 dma_heap
drwxr-xr-x   3 root        root          100 Apr  5 07:38 dri
crw-------   1 root        root      10, 258 Apr  5 07:38 ecryptfs
crw-rw----   1 root        video     29,   0 Apr  5 07:38 fb0
lrwxrwxrwx   1 root        root           13 Apr  5 07:38 fd -> /proc/self/fd
crw-rw-rw-   1 root        root       1,   7 Apr  5 07:38 full
crw-rw-rw-   1 root        root      10, 229 Apr  5 07:38 fuse
crw-------   1 root        root     240,   0 Apr  5 07:38 hidraw0
crw-------   1 root        root      10, 228 Apr  5 07:38 hpet
drwxr-xr-x   2 root        root            0 Apr  5 07:38 hugepages
crw-------   1 root        root      10, 183 Apr  5 07:38 hwrng
crw-------   1 root        root      89,   0 Apr  5 07:38 i2c-0
lrwxrwxrwx   1 root        root           12 Apr  5 07:38 initctl -> /run/initctl
drwxr-xr-x   4 root        root          340 Apr  5 07:38 input
crw-r--r--   1 root        root       1,  11 Apr  5 07:38 kmsg
lrwxrwxrwx   1 root        root           28 Apr  5 07:38 log -> /run/systemd/journal/dev-log
brw-rw----   1 root        disk       7,   0 Apr  5 07:38 loop0
brw-------   1 root        root       7,   1 Apr  5 07:38 loop1
brw-rw----   1 root        disk       7,   2 Apr  5 07:38 loop2
brw-rw----   1 root        disk       7,   3 Apr  5 07:38 loop3
brw-rw----   1 root        disk       7,   4 Apr  5 07:38 loop4
brw-rw----   1 root        disk       7,   5 Apr  5 07:38 loop5
brw-rw----   1 root        disk       7,   6 Apr  5 07:38 loop6
```

In order to identify my terminal, we keep scrolling in the same output until we see the ‘tty’ List.

The ‘tty’ files represent virtual terminals, generated by Linux to allow multiple users or background sessions to run concurrently.

In the ‘ls -l’ output, while the system administrator (root) owns almost all the ‘tty’ devices, the specific line for ‘tty2’ lists the owner as ‘tomerbenb13’ (my user).

This confirms that ‘/dev/tty2’ is our currently active terminal device, and its ownership is dynamically assigned to the logged-in user ‘tomerbenb13’.

We can see that in the screenshot below:

```
crw-rw-rw-   1 root        tty       5,   0 Apr  5 07:38 tty
crw--w----   1 root        tty       4,   0 Apr  5 07:38 tty0
crw--w----   1 root        tty       4,   1 Apr  5 07:38 tty1
crw--w----   1 root        tty       4,  10 Apr  5 07:38 tty10
crw--w----   1 root        tty       4,  11 Apr  5 07:38 tty11
crw--w----   1 root        tty       4,  12 Apr  5 07:38 tty12
crw--w----   1 root        tty       4,  13 Apr  5 07:38 tty13
crw--w----   1 root        tty       4,  14 Apr  5 07:38 tty14
crw--w----   1 root        tty       4,  15 Apr  5 07:38 tty15
crw--w----   1 root        tty       4,  16 Apr  5 07:38 tty16
crw--w----   1 root        tty       4,  17 Apr  5 07:38 tty17
crw--w----   1 root        tty       4,  18 Apr  5 07:38 tty18
crw--w----   1 root        tty       4,  19 Apr  5 07:38 tty19
crw--w----   1 tomerbenb13 tty       4,   2 Apr  5 07:39 tty2
crw--w----   1 root        tty       4,  20 Apr  5 07:38 tty20
crw--w----   1 root        tty       4,  21 Apr  5 07:38 tty21
crw--w----   1 root        tty       4,  22 Apr  5 07:38 tty22
crw--w----   1 root        tty       4,  23 Apr  5 07:38 tty23
crw--w----   1 root        tty       4,  24 Apr  5 07:38 tty24
crw--w----   1 root        tty       4,  25 Apr  5 07:38 tty25
crw--w----   1 root        tty       4,  26 Apr  5 07:38 tty26
crw--w----   1 root        tty       4,  27 Apr  5 07:38 tty27
crw--w----   1 root        tty       4,  28 Apr  5 07:38 tty28
crw--w----   1 root        tty       4,  29 Apr  5 07:38 tty29
crw--w----   1 root        tty       4,   3 Apr  5 07:38 tty3
crw--w----   1 root        tty       4,  30 Apr  5 07:38 tty30
crw--w----   1 root        tty       4,  31 Apr  5 07:38 tty31
crw--w----   1 root        tty       4,  32 Apr  5 07:38 tty32
crw--w----   1 root        tty       4,  33 Apr  5 07:38 tty33
crw--w----   1 root        tty       4,  34 Apr  5 07:38 tty34
crw--w----   1 root        tty       4,  35 Apr  5 07:38 tty35
crw--w----   1 root        tty       4,  36 Apr  5 07:38 tty36
```

4.

Make subdirectories called `work` and `play` under your home folder (~/).

```
carmifr@DESKTOP-81VCQE2:~$ mkdir work
carmifr@DESKTOP-81VCQE2:~$ mkdir play
```

5.

5. Create a file `lab2.txt` and write "hello world" in your **home directory**. *[ write the commands you used into your report]*

```
carmifr@DESKTOP-81VCQE2:~$ nano lab2.txt
carmifr@DESKTOP-81VCQE2:~$ ls
Repos  lab2.txt  lsb_release  play  work
carmifr@DESKTOP-81VCQE2:~$
```

It is also possible to use:

```
tomerbenb13@ubuntu:~$ cd
tomerbenb13@ubuntu:~$ echo "hello world" > lab2.txt
```

Running 'cd' by itself ensures we are starting directly in our home directory.
Then, the 'echo' command generates the required text string.

Now, instead of printing it to the screen, we can use the '>' operator to capture the text and write it directly into a newly created file named 'lab2.txt'.

6.

6. Move `lab2.txt` into the subdirectory `play` *[ write the commands you used into your report]*

The command '**mv**' (move) is used to relocate files or directories from one location to another. In this case, it takes the file 'lab2.txt' from our current location (the home directory) and moves it directly into the 'play' subdirectory (play/).

```
carmifr@DESKTOP-81VCQE2:~$ mv lab2.txt play/
carmifr@DESKTOP-81VCQE2:~$ ls
Repos  lsb_release  play  work
carmifr@DESKTOP-81VCQE2:~$ cd play
carmifr@DESKTOP-81VCQE2:~/play$ ls
lab2.txt
carmifr@DESKTOP-81VCQE2:~/play$
```

7.

7. Change into subdirectory `play` and create a symbolic (soft) link called `terminal` that points to your tty device. What happens if you try to make a hard link to the tty device? Why? *[ write the commands you used int your report for both scenarios]*

```
carmifr@DESKTOP-81VCQE2:~/play$ tty
/dev/pts/0
carmifr@DESKTOP-81VCQE2:~/play$ ln -s /dev/pts/0 terminal
carmifr@DESKTOP-81VCQE2:~/play$ ls -l
total 4
-rw-r--r-- 1 carmifr carmifr 12 Mar 29 15:48 lab2.txt
lrwxrwxrwx 1 carmifr carmifr 10 Mar 29 16:08 terminal -> /dev/pts/0
carmifr@DESKTOP-81VCQE2:~/play$ ln -h /dev/pts/0 terminal
ln: invalid option -- 'h'
Try 'ln --help' for more information.
carmifr@DESKTOP-81VCQE2:~/play$ ln /dev/pts/1 terminal_hard
ln: failed to create hard link 'terminal_hard' => '/dev/pts/1': Invalid cross-device link
carmifr@DESKTOP-81VCQE2:~/play$
```

```
tomerbenb13@ubuntu:~/play$ ln -s /dev/tty2 terminal
tomerbenb13@ubuntu:~/play$ ln /dev/tty2 hard_terminal
ln: failed to create hard link 'hard_terminal' => '/dev/tty2': Invalid cross-dev
ice link
```

The 'cd' command changes the current location into the 'play' subdirectory. Then, the 'ln -s' command creates a symbolic link named 'terminal' that successfully points to the 'tty' device.
The symbolic link worked because it's just a special file that stores a path to another file.
The hard link did not work because hard links to tty/device files are not allowed in this case; hard links are meant for regular files within the same filesystem and are restricted to special device files.

8.

8. What is the difference between listing the contents of directory play with `ls -l` and `ls -lL`? *[ prove your claim by highlighting a specific output of both commands ]*

```
tomerbenb13@ubuntu:~/play$ ls -l
total 4
-rw-rw-r-- 1 tomerbenb13 tomerbenb13 12 Apr  5 09:54 lab2.txt
lrwxrwxrwx 1 tomerbenb13 tomerbenb13  9 Apr  5 10:11 terminal -> /dev/tty2
tomerbenb13@ubuntu:~/play$ ls -lL
total 4
-rw-rw-r-- 1 tomerbenb13 tomerbenb13   12 Apr  5 09:54 lab2.txt
crw--w---- 1 tomerbenb13 tty          4, 2 Apr  5 07:39 terminal
```

The difference between the commands are shown in the output:

**File Type Indicator (First Character):** ‘ls -l’ displays an ‘l’ (indicating a symbolic link), whereas ‘ls -lL’ displays a ‘c’ (indicating the target is a character-oriented device).
**Group Ownership:** ‘ls -l’ shows the link belongs to the user's group (tomerbenb13), while ‘ls -lL’ shows the target file's actual group (tty).
**Size vs. Device ID:** ‘ls -l’ shows the size of the link file (9 bytes, the length of the string "/dev/tty2"). ‘ls-lL’ replaces this with the major and minor device numbers of the target (4, 2).
**Filename Display:** ‘ls -l’ explicitly displays the link and its target (terminal -> /dev/tty2). ‘ls -lL’ hides the redirection and displays only the name (terminal), treating it as the target file itself.

9.

9. Create a file called `hello.txt` that contains the words "`hello world`". Copy `hello.txt` to `terminal`. What happens? Why? *[ write the commands you used to find it out into your report]*

```
carmifr@DESKTOP-81VCQE2:~$ echo "hello world" > hello.txt
carmifr@DESKTOP-81VCQE2:~$ cp hello.txt terminal
hello world
carmifr@DESKTOP-81VCQE2:~$
```

What happened is that "hello world" appeared in the terminal. This happens because everything in Linux is a file, even the terminal. When we copy into it (“terminal” file is a pointer to it), the words are being copied to the file "terminal" where they appeared.

10.

10. Imagine you were working on a system and someone accidentally deleted the `ls` and `find` commands (`by typing: sudo rm /bin/ls /bin/find`). How could you get a list of the files in the current directory? Try it.

We can use the command ‘echo *’. ‘*‘means everything in the directory. ‘echo’ prints whatever comes after the command, so it simply prints the strings representing the folder names.

Another option is ‘ **printf “%s\n/” *** ’, which is the same, but more organized.

```
tomerbenb13@ubuntu:~$ echo *
Desktop Documents Downloads Music Pictures play Public snap Templates Videos work
tomerbenb13@ubuntu:~$ printf "%s\n/" *
Desktop
/Documents
/Downloads
/Music
/Pictures
/play
/Public
/snap
/Templates
/Videos
/work
```

11.

11. How would you create and then delete a file called "$SHELL"? Try it. . *[ write the commands you used to find it out into your report, note the commands used to create this file also]*

To create and then delete a file named “$SHELL”, we would execute the following commands: **touch ‘$SHELL’**, **rm ‘$SHELL’.**

We would use single quotes (‘ ‘) because the $ symbol is a special character used by the system to reference environment variables.
If I simply typed **touch $SHELL**, the shell would expand it to my actual shell path (such as /bin/bash). By enclosing the filename in single quotes, we instruct the shell to strip the special meaning of the $ and treat every character exactly as written.

```
tomerbenb13@ubuntu:~$ touch '$SHELL'
tomerbenb13@ubuntu:~$ ls
'$SHELL'   Documents   Music      play     snap        Videos
 Desktop   Downloads   Pictures   Public   Templates   work
tomerbenb13@ubuntu:~$ rm '$SHELL'
tomerbenb13@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  play  Public  snap  Templates  Videos  work
```

12.

12. How would you create and then delete a file that begins with the symbol #? Try it. . *[ write the commands you used to find it out into your report]*

In Unix shells, "#" signifies the start of a comment. To create a new file starting with "", we will do the following:

```
carmifr@DESKTOP-81VCQE2:~$ touch \#file
carmifr@DESKTOP-81VCQE2:~$ ls
'#file'   Repos   hello.txt   lsb_release   play   terminal   work
carmifr@DESKTOP-81VCQE2:~$ rm \#file
carmifr@DESKTOP-81VCQE2:~$ ls
Repos  hello.txt  lsb_release  play  terminal  work
carmifr@DESKTOP-81VCQE2:~$
```

This way, Linux will know how to add "#" to the file name.

13.

13. How would you create and then delete a file that begins with the symbol -? Try it. . *[ write the commands you used to find it out into your report]*

In UNIX systems, a hyphen (**-**) at the beginning of a word is reserved to indicate command flags or options (such as the '-l ' in 'ls -l'). If we simply typed **touch -myfile**, the system would mistakenly think we are trying to pass the options -m, -y, -f, etc., to the **touch** command, resulting in an "invalid option" error.

By prepending the filename with **./** (which translates to "inside the current directory"), the string no longer begins with a hyphen. This forces the shell to treat **./-myfile** strictly as a file path rather than a list of command options.

```
tomerbenb13@ubuntu:~$ touch -myfile
touch: invalid option -- 'y'
Try 'touch --help' for more information.
tomerbenb13@ubuntu:~$ touch ./-myfile
tomerbenb13@ubuntu:~$ ls
Desktop    Downloads  -myfile  play    snap       Videos
Documents  Music      Pictures Public  Templates  work
tomerbenb13@ubuntu:~$ rm ./-myfile
tomerbenb13@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  play  Public  snap  Templates  Videos  work
tomerbenb13@ubuntu:~$
```

14.

14. Still in your home directory, copy the entire directory `play` to a directory called `work`, preserving the symbolic link.

```
carmifr@DESKTOP-81VCQE2:~$ cp -a play work
carmifr@DESKTOP-81VCQE2:~$ cd work
carmifr@DESKTOP-81VCQE2:~/work$ ls
play
carmifr@DESKTOP-81VCQE2:~/work$ cd play
carmifr@DESKTOP-81VCQE2:~/work/play$ ls
hello.txt  lab2.txt  terminal
carmifr@DESKTOP-81VCQE2:~/work/play$ ls -l terminal
lrwxrwxrwx 1 carmifr carmifr 10 Mar 29 16:08 terminal -> /dev/pts/0
carmifr@DESKTOP-81VCQE2:~/work/play$
```

15.

15. Delete the `work` directory and its contents with one command. Accept no complaints or queries. . *[ write the commands you used to find it out into your report]*

```
carmifr@DESKTOP-81VCQE2:~$ rm -rf work
carmifr@DESKTOP-81VCQE2:~$ ls
Repos  hello.txt  lsb_release  play  terminal
```

rm – short of remove.

-r – (Recursive): This tells Linux to go inside the folder and delete everything - all subfolders and all files. Without this flag, rm cannot delete a directory; it will return an error.

-f – (Force) This tells the system to ignore non-existent files and never ask for confirmation.

16.

16. Experiment with the options on the `ls` command. What do the -d, -R, -F, -i options do?

**ls -d my_folder:** Instead of listing the contents of the folder, this command listed only the directory itself, confirming its existence.

**ls -R**: This triggered a recursive list that mapped out the entire current directory, then automatically dug into every single subdirectory (like **play** and **snap**) to list their contents as well.

**ls -F:** This classifies your files visually. We can see it added a trailing slash (/) to all our directories (like **Desktop/** and **my_folder/**) to distinguish them from standard files (like **my_file**).

**ls -i:** This revealed the underlying file system architecture by printing the unique index number on the far left of every single item (e.g., **1573573 my_file**)

The screenshots are below:

```
tomerbenb13@ubuntu:~$ mkdir my_folder
tomerbenb13@ubuntu:~$ touch my_file
tomerbenb13@ubuntu:~$ ls -d my_folder
my_folder
tomerbenb13@ubuntu:~$ ls -R
.:
Desktop    Downloads  my_file    Pictures  Public  Templates  work
Documents  Music      my_folder  play      snap    Videos

./Desktop:

./Documents:

./Downloads:

./Music:

./my_folder:

./Pictures:

./play:
hello.txt  lab2.txt  terminal

./Public:

./snap:
firmware-updater  snapd-desktop-integration

./snap/firmware-updater:
210  common  current

./snap/firmware-updater/210:

./snap/firmware-updater/common:

./snap/snapd-desktop-integration:
343  common  current
```

```
./snap/snapd-desktop-integration/343:
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos

./snap/snapd-desktop-integration/343/Desktop:

./snap/snapd-desktop-integration/343/Documents:

./snap/snapd-desktop-integration/343/Downloads:

./snap/snapd-desktop-integration/343/Music:

./snap/snapd-desktop-integration/343/Pictures:

./snap/snapd-desktop-integration/343/Public:

./snap/snapd-desktop-integration/343/Templates:

./snap/snapd-desktop-integration/343/Videos:

./snap/snapd-desktop-integration/common:

./Templates:

./Videos:

./work:
tomerbenb13@ubuntu:~$ ls -F
Desktop/    Downloads/  my_file     Pictures/  Public/  Templates/  work/
Documents/  Music/      my_folder/  play/      snap/    Videos/
tomerbenb13@ubuntu:~$ ls -i
1572944 Desktop    1573573 my_file    1572947 Public     1573411 work
1572948 Documents  1573569 my_folder  1572876 snap
1572945 Downloads  1572950 Pictures   1572946 Templates
1572949 Music      1573568 play       1572951 Videos
```